Assignment 3: To develop a distributed system, to find the minimum number of N elements in an array by distributing N/n elements to n number of processors MPI or OpenMP. Demonstrate by displaying the intermediate minimum element calculated at different processors.

```
#include <stdio.h>
#include "mpi.h"

int main(int argc, char* argv[])
{
   int rank, size;
   int num[20];   // N = 20

   MPI_Init(&argc, &argv);
   MPI_Comm_rank(MPI_COMM_WORLD, &rank);
   MPI_Comm_size(MPI_COMM_WORLD, &size);

   // Initialize array
   for(int i = 0; i < 20; i++)
      num[i] = i + 1;

   if(rank == 0)
   {
      int s[4];
      printf("Distribution at rank %d\n", rank);

      // Send N/n elements (20/4 = 5)
      for(int i = 1; i < 4; i++)
         MPI_Send(&num[i * 5], 5, MPI_INT, i, 1, MPI_COMM_WORLD);

      int local_max = num[0];

      for(int i = 1; i < 5; i++)
      {
         if(num[i] > local_max)
            local_max = num[i];
      }

      // Receive from other processes
      for(int i = 1; i < 4; i++)
      {
         MPI_Recv(&s[i], 1, MPI_INT, i, 1, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
      }

      printf("Local max at rank %d is %d\n", rank, local_max);

      int final_max = local_max;

      for(int i = 1; i < 4; i++)
      {
         if(s[i] > final_max)
            final_max = s[i];
      }

      printf("Final Maximum = %d\n\n", final_max);
   }
   else
   {
      int k[5];
      MPI_Recv(k, 5, MPI_INT, 0, 1, MPI_COMM_WORLD, MPI_STATUS_IGNORE);

      int local_max = k[0];

      for(int i = 1; i < 5; i++)
      {
         if(k[i] > local_max)
            local_max = k[i];
      }
```

```
        printf("Local max at rank %d is %d\n", rank, local_max);

        MPI_Send(&local_max, 1, MPI_INT, 0, 1, MPI_COMM_WORLD);
    }

    MPI_Finalize();
    return 0;
}
```

Output:

```
Distribution at rank 0
Local max at rank 1 is 10
Local max at rank 2 is 15
Local max at rank 3 is 20
Local max at rank 0 is 5
Final Maximum = 20
```

Assignment 1: Implement multi-threaded client/server Process communication using RMI.
Find sum of squares of 2 numbers

Client.java

```
import java.rmi.*;
import java.util.Scanner;

public class Client {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      try {
         String serverUrl = "rmi://localhost/Server";
         ServerInterface serverIntf = (ServerInterface) Naming.lookup(serverUrl);

         System.out.print("Enter first number: ");
         double a = sc.nextDouble();

         System.out.print("Enter second number: ");
         double b = sc.nextDouble();

         double result = serverIntf.addSquares(a, b);

         System.out.println("Result (a^2 + b^2) = " + result);

      } catch(Exception e) {
         System.out.println("Exception at Client: " + e);
      }
   }
}
```

Server.java

```
import java.rmi.*;

public class Server {
   public static void main(String[] args) {
      try {
         ServerImplementation serverImpl = new ServerImplementation();
         Naming.rebind("Server", serverImpl);
         System.out.println("Server is Ready...");
      } catch(Exception e) {
         System.out.println("Exception at Server: " + e);
      }
   }
}
```

ServerInterface.java

```
import java.rmi.*;

interface ServerInterface extends Remote {
   public double addSquares(double a, double b) throws RemoteException;
}
```

ServerImplementation

```
import java.rmi.*;
import java.rmi.server.*;

public class ServerImplementation extends UnicastRemoteObject implements ServerInterface {

   public ServerImplementation() throws RemoteException { }

   public double addSquares(double a, double b) throws RemoteException {
      return (a * a) + (b * b);
   }
}
```

OUTPUT

Terminal 1
C:\Users\chet0\OneDrive\Desktop\LP V\A1>rmiregistry
Terminal 2
C:\Users\chet0\OneDrive\Desktop\LP V\A1>java Server
Server is Ready...
Terminal 3
C:\Users\chet0\OneDrive\Desktop\LP V\A1>java Client
Enter first number: 86
Enter second number: 2
Sum of Squares of 86.0 and 2.0 is: 7400.0

Assignment 2 : Develop any distributed application using CORBA to demonstrate object brokering. (Calculator Operations)

Client.java

```
import java.io.BufferedReader;
import java.io.IOException;
import java.io.InputStreamReader;
import CalcApp.*;
import CalcApp.CalcPackage.DivisionByZero;
import org.omg.CosNaming.*;
import org.omg.CosNaming.NamingContextPackage.*;
import org.omg.CORBA.*;
import static java.lang.System.out;

public class CalcClient {
   static Calc calcImpl;
   static BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

   public static void main(String args[]) {

      try {
         ORB orb = ORB.init(args, null);
         org.omg.CORBA.Object objRef = orb.resolve_initial_references("NameService");
         NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef
         String name = "Calc";
         calcImpl = CalcHelper.narrow(ncRef.resolve_str(name));
         while (true) {
            out.println("1. Sum");
            out.println("2. Sub");
            out.println("3. Mul");
            out.println("4. Div");
            out.println("5. exit");
            out.println("--");
            out.println("choice: ");

            try {
               String opt = br.readLine();
               if (opt.equals("5")) {
                  break;
               } else if (opt.equals("1")) {
                  out.println("a+b= " + calcImpl.sum(getFloat("a"), getFloat("b")));
               } else if (opt.equals("2")) {
                  out.println("a-b= " + calcImpl.sub(getFloat("a"), getFloat("b")));
               } else if (opt.equals("3")) {
                  out.println("a*b= " + calcImpl.mul(getFloat("a"), getFloat("b")));
               } else if (opt.equals("4")) {
                  try {
                     out.println("a/b= " + calcImpl.div(getFloat("a"), getFloat("b")));
                  } catch (DivisionByZero de) {
                     out.println("Division by zero!!!");
                  }
               }
            } catch (Exception e) {
               out.println("===");
               out.println("Error with numbers");
               out.println("===");
            }
            out.println("");

         }
         //calcImpl.shutdown();
      } catch (Exception e) {
         System.out.println("ERROR : " + e);
         e.printStackTrace(System.out);
      }
   }
```

```
    static float getFloat(String number) throws Exception {
        out.print(number + ": ");
        return Float.parseFloat(br.readLine());
    }
}
```

Server.java

```
import CalcApp.*;
import CalcApp.CalcPackage.DivisionByZero;

import org.omg.CosNaming.*;
import org.omg.CosNaming.NamingContextPackage.*;
import org.omg.CORBA.*;
import org.omg.PortableServer.*;

import java.util.Properties;

class CalcImpl extends CalcPOA {

    @Override
    public float sum(float a, float b) {
        return a + b;
    }
    @Override
    public float div(float a, float b) throws DivisionByZero {
        if (b == 0) {
            throw new CalcApp.CalcPackage.DivisionByZero();
        } else {
            return a / b;
        }
    }
    @Override
    public float mul(float a, float b) {
        return a * b;
    }
    @Override
    public float sub(float a, float b) {
        return a - b;
    }
    private ORB orb;
    public void setORB(ORB orb_val) {
        orb = orb_val;
    }
}

public class CalcServer {
    public static void main(String args[]) {
        try {
            ORB orb = ORB.init(args, null);
            POA rootpoa = POAHelper.narrow(orb.resolve_initial_references("RootPOA"));
            rootpoa.the_POAManager().activate();
            CalcImpl helloImpl = new CalcImpl();
            helloImpl.setORB(orb);

            org.omg.CORBA.Object ref = rootpoa.servant_to_reference(helloImpl);
            Calc href = CalcHelper.narrow(ref);

            org.omg.CORBA.Object objRef = orb.resolve_initial_references("NameService");
.
            NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef);

            // bind the Object Reference in Naming
            String name = "Calc";
            NameComponent path[] = ncRef.to_name(name);
            ncRef.rebind(path, href);

            System.out.println("Ready..");
```

```
            // wait for invocations from clients
            orb.run();
        } catch (Exception e) {
            System.err.println("ERROR: " + e);
            e.printStackTrace(System.out);
        }

        System.out.println("Exiting ...");

    }
}
```

Output

Terminal 1

```
C:\Users\chet8>cd C:\Users\chete\OneDrive\Desktop\LP V\A2
C:\Users\chete\OneDrive\Desktop\LP V\A2>java CalcServer -ORBInitialPort 1058 -ORBInitialHost localhost
Ready..
```

Terminal 2

```
C:\Users\chet@>cd C:\Users\chet@\OneDrive\Desktop\LP V\A2
C:\Users\chet@\OneDrive\Desktop\LP V\A2>java CalcClient -ORBInitialPort 105
1. Sum
2. Sub
3. Mul
4. Div
5. exit
choice:3
a: 569
b: 87
a+b= 49503.8

1. Sum
2. Sub
3. Mul
4. Div
5. exit
choice:2
a: 89
b: 47
a-b= 42.8

1 Sum
2. Sub
3. Mul
4. Div
5. exit
choice: 4
a: 360
b: 6
a/b= 60.0

1. Sum
2. Sub
3. Mul
4. Div
5. exit
choice:3
a: 360
b: 6
a*b = 216
```

Assignment 4: Implement Berkeley algorithm for clock synchronization.

```
import java.util.*;
class Berkeley {
   public static void synchronize(List<Integer> clocks) {
      int n = clocks.size();
      int sum = 0;
      for (int time : clocks) {
         sum += time;
      }
      int avg = sum / n;
      System.out.println("\nAverage Time: " + avg);
      // Step 3: Calculate adjustments and apply
      System.out.println("\nAdjustments:");
      for (int i = 0; i < n; i++) {
         int current = clocks.get(i);
         int adjustment = avg - current;
         System.out.println("Process " + i + " adjustment = " + adjustment);
         clocks.set(i, current + adjustment);
      }
   }
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter number of clocks: ");
      int n = sc.nextInt();
      List<Integer> clocks = new ArrayList<>();
      System.out.println("Enter clock times:");
      for (int i = 0; i < n; i++) {
         clocks.add(sc.nextInt());
      }
      System.out.println("\nInitial Clock Times:");
      for (int t : clocks) {
         System.out.println(t);
      }
      // Apply Berkeley Algorithm
      synchronize(clocks);
      System.out.println("\nSynchronized Clock Times:");
      for (int t : clocks) {
         System.out.println(t);
      }
   }
}
```

Output

```
C:\Users\chet0\OneDrive\Desktop\LP V\A4>java Berkeley
Enter number of clocks: 3
Enter clock times:
200
50
200

Initial Clock Times:
200
50
200
Average Time: 150
Adjustments:
Process 0 adjustment = -50
Process 1 adjustment = 100
Process 2 adjustment = -50

Synchronized Clock Times:
150
150
150
```

Assignment 5 : Implement token ring based mutual exclusion algorithm.

```
import java.util.*;
class tokenring {
   public static void main(String args[]) {
      Scanner scan = new Scanner(System.in);
      System.out.println("Enter the num of nodes:");
      int n = scan.nextInt();
      int token = 0;
      int ch;
      System.out.println("Ring:");
      for (int i = 0; i < n; i++) {
         System.out.print(" " + i);
      }
      System.out.println(" " + 0);
      do {
         System.out.println("\nCurrent Token at: " + token);
         System.out.println("Enter sender:");
         int s = scan.nextInt();
         System.out.println("Enter receiver:");
         int r = scan.nextInt();
         System.out.println("Enter Data:");
         int data = scan.nextInt();
         System.out.print("Token passing: ");
         while (token != s) {
            System.out.print(token + " -> ");
            token = (token + 1) % n;
         }
         System.out.println(s);
         System.out.println("Sender " + s + " sending data: " + data);
         // Data forwarding
         int temp = s;
         while (temp != r) {
            temp = (temp + 1) % n;
            System.out.println("Data " + data + " forwarded by " + temp);
         }
         System.out.println("Receiver " + r + " received data: " + data);
         token = (r + 1) % n;
         System.out.print("\nSend again? (1-Yes / 0-No): ");
         ch = scan.nextInt();
      } while (ch == 1);
      scan.close();
   }
}
```

Output

```
C:\Users\chet0\OneDrive\Desktop\LP V\A5>java tokenring
Enter the num of nodes:
4
Ring:
 0 1 2 3 0

Current Token at: 0
Enter sender:
2
Enter receiver:
1
Enter Data:
55
Token passing: 0 -> 1 -> 2
Sender 2 sending data: 55
Data 55 forwarded by 3
Data 55 forwarded by 0
Data 55 forwarded by 1
Receiver 1 received data: 55

Send again? (1-Yes / 0-No): 0
```

Assignment 6: Implement Bully and Ring algorithm for leader election.

Ring.java

```
import java.util.Scanner;

public class Ring {
   public static void main(String[] args) {
      Scanner in = new Scanner(System.in);
      System.out.println("Enter the number of processes: ");
      int num = in.nextInt();

      Rr[] proc = new Rr[num];
for (int i = 0; i < num; i++) {
         proc[i] = new Rr();
         proc[i].index = i;
         System.out.println("Enter the ID of process " + (i + 1) + ": ");
         proc[i].id = in.nextInt();
         proc[i].state = "active";
         proc[i].f = 0;
      }
      for (int i = 0; i < num - 1; i++) {
         for (int j = 0; j < num - 1; j++) {
            if (proc[j].id > proc[j + 1].id) {
               Rr temp = proc[j];
               proc[j] = proc[j + 1];
               proc[j + 1] = temp;
            }
         }
      }
      for (int i = 0; i < num; i++) {
         System.out.print("[" + i + "] " + proc[i].id + " ");
      }
      proc[num - 1].state = "inactive";
      System.out.println("\nProcess " + proc[num - 1].id + " selected as coordinator");
      while (true) {
         System.out.println("\n1. Election\n2. Quit");
         int ch = in.nextInt();

         // Reset flags
         for (int i = 0; i < num; i++) {
            proc[i].f = 0;
         }

         switch (ch) {
            case 1:
               System.out.println("Enter the process number that initializes the election: ");
               int init = in.nextInt();
               int temp2 = init;
               int temp1 = init + 1;
               int i = 0;

               while (temp2 != temp1) {
                  if (temp1 == num) {
                     temp1 = 0;
                  }
                  if ("active".equals(proc[temp1].state) && proc[temp1].f == 0) {
                     System.out.println("Process " + proc[init].id + " sends a message to Process " + proc[temp1].id);
                     proc[temp1].f = 1;
                     init = temp1;
                     i++;
                  }
                  temp1++;
               }

               System.out.println("Process " + proc[init].id + " sends a message to Process " + proc[temp1].id);
               int max = -1;
```

```
                int m = -1;
for (int j = 0; j < num; j++) {
   if (proc[j].id > m) {
      m= proc[j].id;
   }
}
                System.out.println("Process " + m + " selected as coordinator");
                for (int k = 0; k < num; k++) {
                   if (proc[k].id == m) {
                      proc[k].state = "inactive";
                   }
                }
                break;
             case 2:
                System.out.println("Program terminated.");
                in.close();
                return;
             default:
                System.out.println("Invalid response.");
                break;
          }
       }
   }
}

class Rr {
   public int index; // To store the index of the process
   public int id; // To store the ID/name of the process
   public int f;
   public String state; // Indicates whether the process is active or inactive
}
```

Output

```
C:\Users\chet0\OneDrive\Desktop\LP V\A6>java Ring
Enter the number of processes:
4
Enter the ID of process 1:
2
Enter the ID of process 2:
3
Enter the ID of process 3:
4
Enter the ID of process 4:
5
[0] 2 [1] 3 [2] 4 [3] 5
Process 5 selected as coordinator

1. Election
2. Quit
1
Enter the process number that initializes the election:
3
Process 5 sends a message to Process 2
Process 2 sends a message to Process 3
Process 3 sends a message to Process 4
Process 4 sends a message to Process 5
Process 5 selected as coordinator

1. Election
2. Quit
```

Bully.java

```
import java.util.Scanner;

public class Bully {
   static boolean[] state = new boolean[5];
   public static int coordinator = 4;
   public static void getStatus() {
      System.out.println("\n+------Current System State-----+");
      for (int i = 0; i < state.length; i++) {
         System.out.println("| P" + (i + 1) + ":\t" + (state[i] ? "UP" : "DOWN") +
               (coordinator == i ? "\t<-- COORDINATOR\t|" : "\t\t\t|"));
      }
      System.out.println("+--------------------------------+");
   }
   public static void up(int up) {
      if (state[up - 1]) {
         System.out.println("Process " + up + " is already up");
      } else {
         state[up - 1] = true;
         System.out.println("--------Process " + up + " held election-------");
         for (int i = up; i < state.length; ++i) {
            System.out.println("Election message sent from process " + up + " to process " + (i + 1));
         }
         for (int i = state.length - 1; i >= 0; --i) {
            if (state[i]) {
               coordinator = i;
               break;
            }
         }
      }
   }

   public static void down(int down) {
      if (!state[down - 1]) {
         System.out.println("Process " + down + " is already down.");
      } else {
         state[down - 1] = false;
         if (coordinator == down - 1) {
            setCoordinator();
         }
      }
   }


   public static void mess(int mess) {
      if (state[mess - 1]) {
         if (state[coordinator]) {
            System.out.println("Message Sent: Coordinator is alive");
         } else {
            System.out.println("Coordinator is down");
            System.out.println("Process " + mess + " initiated election");
            for (int i = mess; i < state.length; ++i) {
               System.out.println("Election sent from process " + mess + " to process " + (i + 1));
            }
            setCoordinator();
         }
      } else {
         System.out.println("Process " + mess + " is down");
      }
   }

   public static void setCoordinator() {
      for (int i = state.length - 1; i >= 0; i--) {
         if (state[i]) {
            coordinator = i;
            break;
         }
      }
```

```
    }

    public static void main(String[] args) {
        int choice;
        Scanner sc = new Scanner(System.in);
        for (int i = 0; i < state.length; ++i) {
            state[i] = true;
        }
        getStatus();
        do {
            System.out.println("+.....MENU.....+");
            System.out.println("1. Activate a process.");
            System.out.println("2. Deactivate a process.");
            System.out.println("3. Send a message.");
            System.out.println("4. Exit.");
            System.out.println("+.....+");
            choice = sc.nextInt();
            switch (choice) {
                case 1: {
                    System.out.println("Activate process:");
                    int up = sc.nextInt();
                    if (up == 5) {
                        System.out.println("Process 5 is the coordinator");
                        state[4] = true;
                        coordinator = 4;
                        break;
                    }
                    up(up);
                    break;
                }
                case 2: {
                    System.out.println("Deactivate process:");
                    int down = sc.nextInt();
                    down(down);
                    break;
                }
                case 3: {
                    System.out.println("Send message from process:");
                    int mess = sc.nextInt();
                    mess(mess);
                    break;
                }
            }
            getStatus();
        } while (choice != 4);
        sc.close();
    }
}
```

Output

```
C:\Users\chet0\OneDrive\Desktop\LP V\A6>java Bully

+------Current System State-----+
| P1:   UP                     |
| P2:   UP                     |
| P3:   UP                     |
| P4:   UP                     |
| P5:   UP      <-- COORDINATOR |
+------------------------------+
+.....MENU.....+
1. Activate a process.
2. Deactivate a process.
3. Send a message.
4. Exit.
+.....+
2
Deactivate process:
```

```
5

+------Current System State-----+
| P1:   UP                     |
| P2:   UP                     |
| P3:   UP                     |
| P4:   UP     <-- COORDINATOR |
| P5:   DOWN                   |
+------------------------------+
+........MENU........+
1. Activate a process.
2. Deactivate a process.
3. Send a message.
4. Exit.
+.....................+

3
Send message from process:
5
Process 5 is down

+------Current System State-----+
| P1:   UP                     |
| P2:   UP                     |
| P3:   UP                     |
| P4:   UP     <-- COORDINATOR |
| P5:   DOWN                   |
+------------------------------+
+........MENU........+
1. Activate a process.
2. Deactivate a process.
3. Send a message.
4. Exit.
+.....................+
1
Activate process:
5
Process 5 is the coordinator

+------Current System State-----+
| P1:   UP                     |
| P2:   UP                     |
| P3:   UP                     |
| P4:   UP                     |
| P5:   UP     <-- COORDINATOR |
+------------------------------+
+........MENU........+
1. Activate a process.
2. Deactivate a process.
3. Send a message.
4. Exit.
+.....................+
4

+------Current System State-----+
| P1:   UP                     |
| P2:   UP                     |
| P3:   UP                     |
| P4:   UP                     |
| P5:   UP     <-- COORDINATOR |
+------------------------------+
```

Assignment No. 7: Create a simple web service and write any distributed application to consume the web service. – Temperature Converter

Server.py

```
import requests
def convert_temperature():
    value = float(input("Enter temperature value: "))
    unit = input("Enter unit (C/F): ").upper()

    url = 'http://localhost:5000/convert'
    data = {
        'value': value,
        'unit': unit
    }

    response = requests.post(url, json=data)
    result = response.json()

    if 'error' in result:
        print(result['error'])
    else:
        print(f"\n{result['original']}°{result['unit']} = "
              f"{result['converted_value']}°{result['converted_to']}")
convert_temperature()
```

Client.py

```
from flask import Flask, request, jsonify
app = Flask(__name__)
@app.route('/convert', methods=['POST'])
def convert_temp():
    data = request.get_json()
    value = float(data['value'])
    unit = data['unit'].upper()
    if unit == 'C':
        result = (value * 9/5) + 32
        converted_to = 'F'
    elif unit == 'F':
        result = (value - 32) * 5/9
        converted_to = 'C'
    else:
        return jsonify({'error': 'Invalid unit. Use C or F'}), 400
    return jsonify({
        'original': value,
        'unit': unit,
        'converted_value': round(result, 2),
        'converted_to': converted_to
    })
if __name__ == '__main__':
    app.run()
```

Output

Terminal 1

```
C:\Users\chet0\OneDrive\Desktop\LP V\A7>python server.py
 * Serving Flask app 'server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI
server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [24/Apr/2026 13:45:52] "POST /convert HTTP/1.1" 200 -
```

Terminal 2

```
C:\Users\chet0\OneDrive\Desktop\LP V\A7>python app.py
Enter temperature value: 37
Enter unit (C/F): c
Result:37.0°C = 98.6°F
```